GeneticPrism: Multifaceted Visualization of Citation-based Scholarly Research Evolution

Ye Sun, Zipeng Liu, Yuankai Luo, Lei Xia, Lei Shi

***Abstract*—Understanding the evolution of scholarly research is essential for many real-life decision-making processes in academia, such as research planning, frontier exploration, and award selection. Popular platforms like Google Scholar and Web of Science rely on numerical indicators that are too abstract to convey the context and content of scientific research, while most existing visualization approaches on mapping science do not consider the presentation of individual scholars' research evolution using curated self-citation data. This paper builds on our previous work and proposes an integrated pipeline to visualize a scholar's research evolution from multiple topic facets. A novel 3D prism-shaped visual metaphor is introduced as the overview of a scholar's research profile, whilst their scientific evolution on each topic is displayed in a more structured manner. Additional designs by topic chord diagram, streamgraph visualization, and inter-topic flow map, optimized by an elaborate layout algorithm, assist in perceiving the scholar's scientific evolution across topics. A new six-degree-impact glyph metaphor highlights key interdisciplinary works driving the evolution. The proposed visualization methods are evaluated through case studies analyzing the careers of prestigious Turing award laureates, one major visualization venue, and a focused user study. The GeneticPrism web application is publicly available at https://genetic-flow.com.**

***Index Terms*—Academic networks, multifaceted visualization, scientific research evolution**

I. Introduction

Understanding and analyzing the evolution of scholarly scientific research is crucial for comprehending the intricate nature of scholarly contribution and influence. Many real-world tasks such as award selection, research topic planning [43] and frontier exploration [23], as well as tenure evaluation [48] could benefit from such study. Visualization is indispensable in delineating the evolution of scientific topics at the scholar level [60] [55], institutional scale [30] [52], and for entire research fields [42]. Compared to abstract indicators like h-index and impact factor popular in scholarly platforms (Google Scholar [27], Web of Science [15], etc.), the visualization of collaboration networks [58], co-citation/bibliographic graphs [31], and topic content evolutions [30] analyze contextual data better, uncover hidden scientific patterns, and provide detailed reasoning.

In the literature, mapping the evolution of sciences has been the research community's goal ever since Eugene Garfield and others established the famous Institute for Scientific Information decades ago based on citation data [26]. Later on, CiteSpace (II) pioneered the visualization of research fronts and intellectual base using the concept of co-citations [13]. Nature also launched a visualization project recently to mark the journal's 150th anniversary, illustrating 19 million articles on their co-citation networks [42]. Till now, numerous systems and software have been proposed for mapping science, using a variety of information visualization techniques such as networks [10], maps [24], matrices [55], etc. Our work builds over the latest GeneticFlow design by Xiao and Shi [60], who utilize curated self-citation data to illustrate the scientific research evolution of individual scholars, a task that very few existing visualization approaches consider.

The GeneticFlow platform has been extended to support the visual profiling of more than 100k scholars from 12 research areas of computer science [6]. The technique works well on most scholars with a small to medium number of publications in their entire career. However, for top scholars accumulating hundreds of papers and thousands of self-citation links, GeneticFlow adopting classical visualization algorithms suffers from huge visual clutter and edge crossings, a known issue for large graph visualization. Moreover, the original design improvises the display of the most crucial topic information with a plain color-coding on paper nodes. Both intra-topic and inter-topic research evolutions are hardly perceived due to the complex nature of topic distribution over a scholar's career.

In this work, we are motivated by the observation of a universal pattern in scholars' scientific evolution: their works on the same research topic are normally more interconnected (by citation links) than those on different topics. A straightforward proposal would be splitting a top scholar's GeneticFlow graph into multiple topic-based facets and displaying them separately. Visualizing each topic facet with a much smaller number of papers and citations in a standalone view will be more effective and intuitive for analysis. This simple approach, though compelling, still faces multiple technical challenges. First, how can we visually accommodate a scholar's potentially 10+ topic facets in a traditional 2D display space? Second, while inter-topic citations are more infrequent, they are at least as crucial as intra-topic citations because the key information of topic evolution is encapsulated there. How do we visualize a scholar's scientific evolution in a topic-based manner while still uncovering topic-level interactions? Third, the research topic development of a scholar is not linear in that many works overlap accross multiple topics. How do we display this critical evidence of a scholar's scientific evolution? We make the following contribution to tackle the challenges above.

- We solve the problem of analyzing and visualizing

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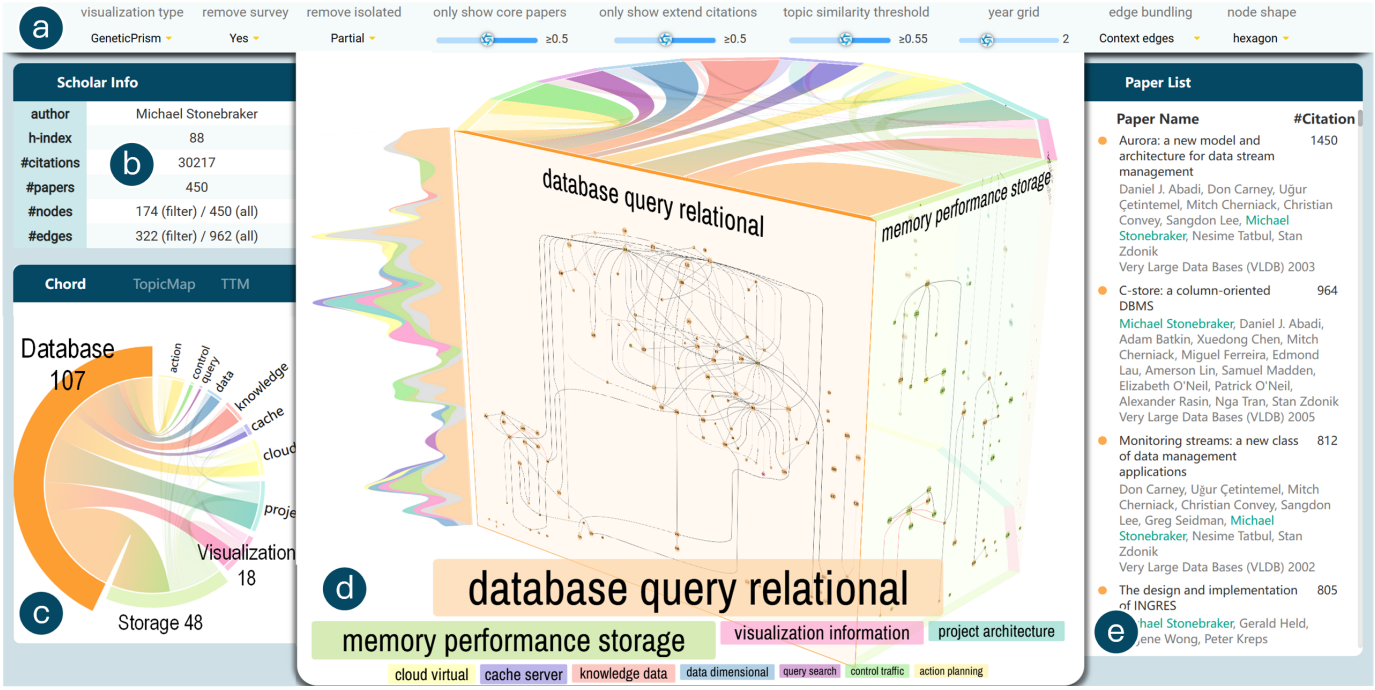


Fig. 1: GeneticPrism interface: (a) control panel; (b) scholar demographics and GeneticFlow graph statistics; (c) chord diagram to display inter-topic citation influence interactions; (d) main GeneticPrism (current) or GeneticScroll designs visualizing Shirato graphs; (e) paper list and detailed info panel.

multiple topic facets. Case studies on two prestigious Turing Award laureates demonstrate the effectiveness of GeneticPrism by clearly delineating their major research threads across multiple topic facets, visually reasoning on important topic evolution patterns, and identifying key interdisciplinary papers that drive their research evolution. A user study with 20 participants evaluated GeneticPrism on 7 questions, demonstrating huge improvements in task performance, interaction, topic modeling effectiveness, and user engagement.

- To address the challenge of multiple topic facets, we introduce GeneticPrism (Figure 1(d)), a novel 3D prism-shaped visual metaphor, to integrate multiple topic facets of a scholar in the same display. Complemental designs, including a topic chord diagram (Figure 1(c)), customized animations and interactions, are also presented to illuminate the overview of a scholar's scientific research evolution. We also propose an end-to-end data analysis and topic modeling pipeline to support GeneticPrism visualization (Figure 2);
- To support interdisciplinary analysis, we introduce the GeneticScroll design, which orchestrates a hierarchical graph visualization layer for intra-topic research evolution with a flow map and streamgraph visualization layer on inter-topic citation influences. An elaborate graph layout algorithm is proposed to coordinate the two layers and minimize edge crossings and visual clutter. We also propose a new six-degree-impact glyph design highlighting interdisciplinary works on GeneticScroll. Note that throughout this work, by interdisciplinary, we mean overlaps among fine-grained research topics, not using its original definition among high-level research areas;

II. Related Work

A. Visualizing Scholarly Research Data

Visualizing scholarly research data involves understanding and presenting a scholar's contributions over time and examining how topics, authors, and scientific trends develop and impact their fields, in which four key features provide valuable insights. "Multiple topic facet" tracks research trends in different topic facets, revealing interaction and emerging areas. "Temporal visualization" highlights the progression of research topics and productivity over time. "Venue analysis" examines publications in a journal or conference, uncovering milestones and evolution. "Citation-based evolution" maps the citation data into research evolution, identifying key outlets and cross-disciplinary activities. A summary of related works and comparisons is provided in Table I, categorized into three types: publication list visualization, collaboration network visualization, and visual analysis of citation-based profile.

Publication list visualization focuses on presenting demographics and analyzing multiple factors influencing academic careers. ACSeeker [54] explores potential factors of academic success across different career stages. ImpactVis [55] uses a matrix form to visualize individual careers and domain trends. Vis Author Profiles [34] combines text and visualization for insights into individual and domain impacts. ScholarPlot [37] and Pathway [57] use scatter plots and area charts to depict career trajectories. However, these works lack analysis of relationships between publications, provide limited research evolution insights, and perform only basic topic analysis, lacking in-depth exploration.

Collaboration network visualization focuses on linking publications through co-authorship to reveal collaboration pat-

TABLE I: A comparative analysis of related works on visualizing scientific impact evolution. The final row gives GeneticPrism, our proposed method. ✓: Yes, ○: Partial.

Method	Multiple topic facet	Temporal visualization	Venue analysis	Citation-based evolution	Visual technique	Year
GeneticFlow [60]	✓	✓		✓	node-link	2023
PubExplorer [61]	○	✓	✓	○	state map, histogram	2023
ACSeeker [54]	○	✓			timeline, matrix	2022
VisualBib [17]	○	✓	✓	○	node-link, radar, streamgraph	2022
Eiffel [31]	○	✓	✓	○	Flow map	2020
Nature150 [42]	○	○	○		3D node-link	2019
ImpactVis [55]	○	✓	✓	○	matrix, glyph, stacked bar chart	2018
Vis Author Profiles [34]	○	✓			text, stacked bar chart	2018
CiteRivers [30]	✓	✓	✓	○	streamgraph	2016
ScholarPlot [37]		✓			scatter plot	2016
VEGAS [49]	○		✓	○	node-link	2015
VOSviewer [52]	○		✓	○	node-link	2014
Pathway [57]		✓			area chart	2013
GeneticPrism	✓	✓	✓	✓	**3D prism, node-link, chord, streamgraph, flow map**	**Ours**

terns. EgoSlider [58] uses a timeline-based visualization for evolutionary collaboration networks. EgoLines [62] employs a "subway map" metaphor for temporal dynamics. Reitz [46] uses a scholar's ego-centric node-link diagram for temporal distributions of joint works. MENA [29] represents ego-centric networks as dynamic graphs in small multiples. Fung et al. [25] suggest a botanically inspired tree visualization summarizing collaborations over time. Episogram [12] presents a novel visualization tool for summarizing egocentric social interactions, enabling users to explore and compare social behaviors over time. However, these works focus on co-authorships, while our interest lies in citations.

Visual analysis of citation-based profile treats citation links as representations of academic influence and focuses on analyzing scholars' research profile. Eiffel [31] applies triple summarizations and uses a dynamic flow map for hierarchical and temporal changes of a paper's citation influence. CiteRivers [30] employs a visual streamgraph metaphor to depict thematic developments over time. VEGAS [49] employs visual influence graph summarization to enable the exploration of highly influential papers. The Nature journal [42] offers galaxy and light cone views for co-citation networks. VOSviewer [52] uses clustering and network visualizations to highlight research clusters and citation connections, providing an overview of research landscapes. PubExplorer [61] integrates science maps and citation pattern histograms to reveal research trends and key collaborations, supporting interdisciplinary exploration. VisualBib [16] utilizes a semantic zz-structure and narrative views to dynamically link bibliographic metadata, aiding in bibliographic storytelling. While these approaches analyze citation-based impact and correlations, they either model topic facets separately without considering topic influence and interaction or fail to explicitly display research evolution or trace research lineages, such as fine-grained interdisciplinary activities or research transformations.

GeneticFlow (GF) [36], [60] introduces methods for constructing graph-based scholar profiles, illustrating academic trajectories and scholarly research. This structured representation of a scholar's scientific development emphasizes the evolution of research topics. Still, it does not work well for top scholars with abundant publications and citation influence. Additionally, its intrusive topic modeling, which assigns only one topic per paper, reduces the possibility of understanding domain insight in complex, multi-topic scenarios. In comparison, our approach focuses on scholarly research evolution and trajectories visualization across various dimensions, integrating topic-based, temporal, venue analysis, and research evolution visualizations using a combination of node-link graphs, chord diagrams, streamgraphs, and flow maps.

B. Visualizing Multilayer Networks

Multilayer network visualization, which represents systems where entities interact across multiple contexts or dimensions [18], benefits fields such as biology, sociology, and digital humanities. Nodes can belong to multiple layers, with edges existing within or spanning across layers. Recent surveys [22], [39] highlight the need for advanced visualization techniques to manage the complexity and scale of multilayer networks.

Several innovative approaches have been developed. MuxViz [19] provides a platform for analyzing and visualizing multilayer networks using techniques like adjacency matrices and node-link diagrams. Detangler [47] offers coordinated views to explore inter-layer relationships. Multilayer Graph Edge Bundling [9] addresses edge clutter by grouping edges with common paths. Refinery [33] supports cross-layer connectivity tasks through associative browsing. HybridVis [35] integrates multiple visualization techniques for large multivariate networks, providing a flexible approach to visualize different data aspects. NetworkAnalyst [59] uses chord diagrams for inter-layer relationships in gene expression data. Dynamic Communities [53] visualizes the evolution of community structures within multilayer networks. g-Miner [14] offers interactive techniques for exploring and manipulating graph data, including creating and refining groupings within layers. Shirato et al. [50] enhance the understanding of complex temporal relationships in multivariate time series through advanced visualization techniques across multiple dimensions. GraphDice [7] extends ScatterDice [21] for multivariate networks, providing a consistent interface for visualizing node and edge attributes. Similar to GeneticPrism, GraphDice supports exploring complex multilayer networks through filtering and partitioning based on various attributes but focuses more on interactive exploration of network dimensions and does not present the influence among groups.

The survey by McGee et al. [39] categorizes tasks into cross-layer connectivity, cross-layer entity comparison, layer manipulation, and layer comparison. However, tasks addressing inter-relationships between layers, such as cross-layer flows, are rarely explored. GeneticPrism addresses this gap by visualizing inter-layer relationships through a suite of novel visualization designs. Furthermore, GeneticPrism supports overlapping topic visualization that simultaneously displays nodes appearing in multiple network layers.

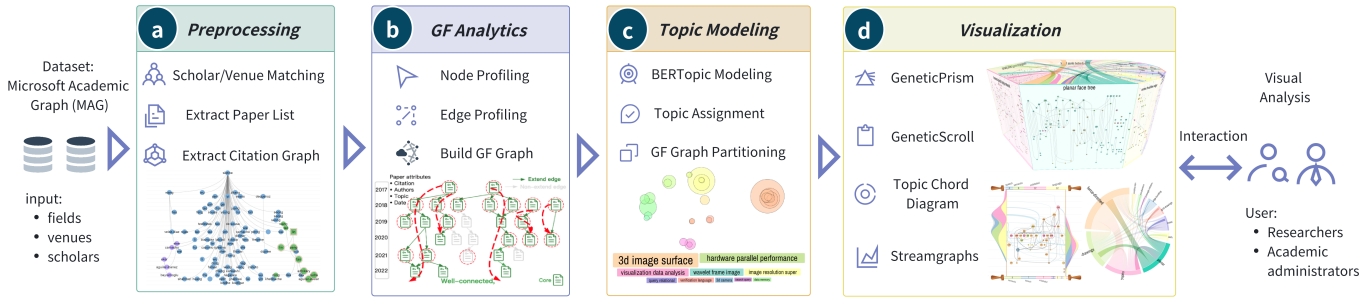


Fig. 2: GeneticPrism pipeline over MAG to illustrate scholarly research evolution: (a) data pre-processing; (b) GF analytics to build citation influence graph; (c) topic modeling for multifaceted analysis; (d) visualization designs.

TABLE II: GF graph statistics used in this work, including award recipients, top scholars in CS subfields, and academic venues.

Category	Name	#Authors	#Topic	#Papers	#Links
Award Recipients	ACM Fellows	1285	324	421205	818548
	Turing Award Winners	67	324	17625	30780
Top Scholars in 12 CS Sub-fields	Graphics & Visualization	11980	105	271418	262957
	Artificial Intelligence	54637	497	4836561	8420229
	(Full list in Appendix)				
Academic Venues	International Symposium on Graph Drawing	N/A	28	1703	3128

III. Scholarly Research Evolution

Topic facets refer to well-established research topics within academic literature. The term multiple topic facets describes the phenomenon in which individual scholars engage in several distinct research topics concurrently throughout their careers. To study this phenomenon, the proposed pipeline to illustrate scholarly research evolution is shown in Figure 2. The raw academic data after preprocessing is first modeled by GF analytics into citation influence graphs, or GF graphs in short. A new topic modeling stage is introduced to assemble multiple topic facets on each GF graph. GeneticPrism and other visual metaphors are then designed for the multi-view visualization.

A. Academic Data Source and Pre-processing

Our primary data source is MAG [41], the largest open academic database nowadays, with over 237 million papers, 240 million authors, and 1.63 billion citations from all research disciplines till Sept. 2021. As shown in Table II, the raw MAG data is pre-processed to obtain the citation graph of academic entities in three categories: award recipients, including Turing Award winners and ACM Fellows, top scholars in 12 computer science (CS) sub-fields (e.g., Graphics & Visualization, Artificial Intelligence), and an academic venue of GD. The full list of 12 CS sub-fields, along with their associated venues and topics, is provided in Appendix 1.1.

For award recipients, lists of awardees were collected from official sources (ACM Fellows [2] and Turing Award recipients [1]). These scholar lists were matched to MAG data based on their names and research fields. For the top scholars in a subfield, we defined a subfield database consisting of papers from specific venues and topics. Scholars were ranked by their h-index calculated from all papers within the subfield, and we selected those above a predefined threshold (e.g., the 11980 scholars in graphics & visualization with h-index above 5). Scholar name matching and disambiguation were conducted to ensure that the correct scholars were identified in our dataset, and all their papers from MAG were extracted. For the GD venue, we simply discover its venue ID in MAG and retrieve all the papers linked to that ID. After obtaining the paper list for each scholar/venue, we fetched all citation links among these papers to build citation networks. The citation networks are then used to build GF graphs and conduct topic modeling.

B. GeneticFlow Graph Model

As shown in Figure 2(b), we implement the GeneticFlow (GF) graph model from [36] to analyze scholarly impact. The model constructs a directed graph from a scholar's self-citation network where node profiling identifies *core papers* where the scholar is first author or advises the first author, while edge profiling identifies *extend-type citations* through supervised classification (detailed in Appendix 1.2).

The resulting GF graph retains papers exceeding a core probability threshold (default 0.5) and citations exceeding an extend-type probability threshold (default 0.5), emphasizing research continuity. Importantly, our visualization system supports real-time adjustment of these threshold parameters, enabling users to interactively switch between the complete self-citation graph and a more refined core evolution graph.

C. Topic Modeling

To resolve the scalability issue in visualizing very large GF graphs, we propose to slice a scholar's GF graph into multiple sub-graphs using topic modeling methods. It is observed that the citation influence graph inside each topic is usually much smaller yet more interpretable, as shown by the case studies in Sec. VI. The missing interactions among GF subgraphs can be displayed by elaborate visualization designs in Sec. IV. Our approach involves three key steps: BERTopic modeling to extract topics, topic assignment to associate papers with topics, and GF graph partitioning into topic-based sub-graphs.

We apply the latest neural topic modeling method, BERTopic [28], to extract topics. The BERTopic pipeline comprises three main steps: using BERT embeddings to represent each document as a dense, high-dimensional vector; applying UMAP projection [40] to map these vectors into a low-dimensional space; and performing HDBSCAN clustering [45] to detect topics from the documents. For each detected topic, a list of keywords and an embedding vector are computed. Notably, a single topic model is trained for all scholars in the same field or award category (e.g., Graphics & Visualization,

corresponding to one row in Table II) and applied consistently across all authors in that domain. During the topic modeling process, only titles and abstracts are used, as full texts are not available in the MAG database.

In the topic assignment step, to associate each paper with multiple topics, we compute the cosine similarity between the paper's embedding and all learned topic embeddings. Each paper is assigned to the top 3 topics for which the similarity exceeds a threshold (e.g. 0.6). Sensitivity analysis of this threshold is detailed in Appendix 1.3. This threshold was empirically chosen based on observations that papers with a similarity above this value have fewer topics and exhibit strong semantic relevance to their assigned topics. However, this value is adjustable within the visualization interface to accommodate different datasets and user preferences.

Finally, the full GF graph of a scholar is partitioned into multiple per-topic subgraphs by selecting all papers associated with a given topic and their internal citation links. Since each paper can belong to multiple topics (up to 3), it may appear as a node in several subgraphs. This segmentation enables an overlapping representation of topics within the GF graph.

To assess the consistency of topic modeling, we compared the highest-similarity topics for each paper across multiple iterations using an adjusted version of Krippendorff's Alpha [38]. Since topic learning involves clustering rather than classification, the learned topic categories and their order may vary between iterations. Therefore, label alignment using the Hungarian algorithm was performed before calculating K-alpha. We conducted repeated experiments on two datasets: ACM Fellow (259,526 papers) and Graphics & Visualization (130,650 papers), yielding K-alpha scores of 0.738 and 0.944, respectively. These results demonstrate the high consistency and reliability of the topic modeling process.

D. User and Task Characterization

Building over faceted GF graphs, GeneticPrism visualizations are designed to effectively delineate the scientific research evolution of key scholars and academic venues. Our technique targets two types of users:

- **Researchers** who aim to analyze the academic development of themselves and other scholars from a data-driven perspective. The lessons learned and patterns detected can potentially assist them in future topic planning, collaboration building, and research frontier understanding;
- **Academic administrators** with regular responsibility for technology recruitment, award evaluation, award/project selection, or even agenda-setting for a research field. The quantitative approach by GeneticPrism serves as a nice complement to the traditional subjective peer reviews.

To fulfill the requirements of these users, GeneticPrism supports the following tasks:

- **T1: temporal and topical overview of a scholar's research.** On the first hand, most users will need an overall understanding of a scholar's research, including the research contribution across topics and over time. The design should allow a macroscopic view with a time dimension that can juxtapose and compare the scholar's research evolution on multiple topics;
- **T2: detailed analysis of a scholar's research evolution on a single research topic.** For any specific topic, e.g., the one that the user is currently working on or the administrator is overseeing, s/he will need to drill down to details to complete their job. The low-level tasks include but are not limited to identifying key papers, understanding research threads/clusters, and predicting topical/statistical trends;
- **T3: discover influence and interaction patterns among multiple research topics of a scholar.** For topic planning and research outlook, users will need to figure out the causal relationship among multiple research topics of a scholar. This can be achieved through the visual analysis of citation influence patterns among papers belonging to these topics;
- **T4: identifying key interdisciplinary papers and their influence patterns.** During topic transitions of a scholar, there usually is not a clear-cut pattern between old and new topics. It is crucial for our users to identify the key papers that conduct interdisciplinary research on these transitional topics. The context of this research can help explain the reasoning behind the process of research evolution.

IV. Visualization

Over GF analytics and topic modeling to illustrate the research evolution of key scholars and academic venues. As shown in Figure 1, the interface is composed of five panels: a visualization controller to config the entire system (Figure 1(a)), a demographics panel to display statistics of the current scholar (Figure 1(b)), a topic interaction panel to present high-level citation influence patterns among topics (Figure 1(c)), a paper list panel to detail high-impact papers of that scholar (Figure 1(e)), and finally the main panel in the middle to visualize GF graphs. In the default overview mode, the GeneticPrism visualization is displayed (Figure 1(d)); in the other per-topic mode, the main panel will switch to the GeneticScroll visualization (Figure 4).

A. GeneticPrism

In a default mode, the main visualization panel of our system features the GeneticPrism design, which fulfills the T1 overview task defined above. As shown in Figure 3(a), a 3D prism-shaped metaphor is adopted, which consists of two primary components for visualization:

- Top view (horizontal face, shown in Figure 3(b)): This provides a snapshot of the topic chord diagram, offering an overview of the inter-topic influence (T3). Specifically, we directly use the inscribed polygon corresponding to the chord diagram to ensure flat vertical faces.
- Side views (multiple vertical faces, shown in Figure 3(c)): Each side view represents a per-topic GF subgraph (T2). All side views share the same height, as the vertical axis corresponds to time (publication year). The width of each side view is determined by the angle of the topic in the chord diagram. More influential topics occupy a larger angular and correspond to wider side views.

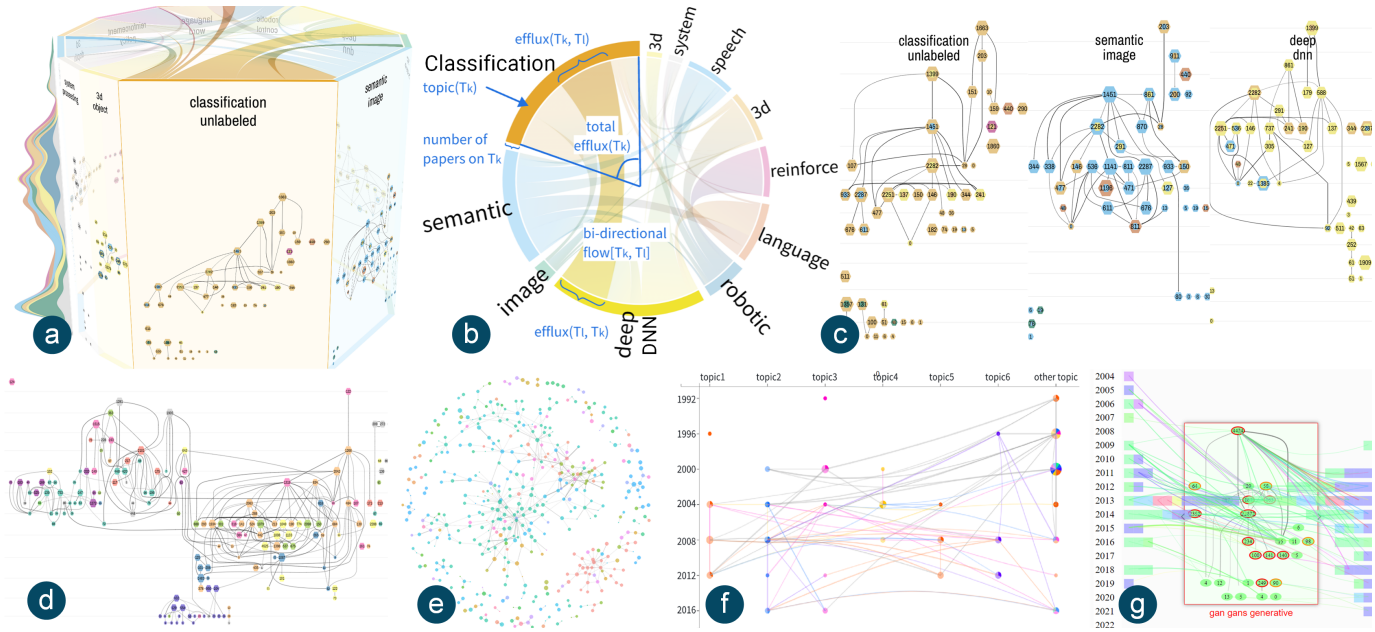


Fig. 3: GeneticPrism design: (a) GeneticPrism visualization of scholar Andrew Y. Ng; (b) top view (topic chord diagram) showing the relationships between topics; (c) side view visualizing the per-topic GF subgraphs on three topics of the scholar; (d-g) alternative design: full GF graph, force-directed layout, temporal-topical flow matrix, GFsubgraph with stacked bar charts.

The entire visualization is rendered in semi-transparent colors so that several topic-based GF subgraphs can be displayed simultaneously for comparison purposes. From another perspective, the GeneticPrism metaphor is designed to mimic a cultural monument, symbolizing the multifaceted and impactful scientific research contributions of prestigious scholars.

Topic chord diagram (Figure 3(b)) is introduced to visualize inter-topic interactions for the T3 task, and it is presented both as the top view of GeneticPrism and as a standalone view in the interface (Figure 1(c)). The chord diagram illustrates the citation influence graph among all derived research topics of a scholar, denoted as $T_1, T_2, \ldots, T_N$. The directed flow from topic T_k to topic T_l represents the citation influence from T_k to T_l, with the efflux calculated as the total number of citation influence links from T_k to T_l:

$$\begin{aligned}\text{efflux}(T_k, T_l) = |\{e_{ij} \mid e_{ij} \in E, v_i \in V(T_k) \wedge \\ v_i \notin V(T_l) \wedge v_j \in V(T_l)\}|\end{aligned} \quad (1)$$

where $|\cdot|$ denotes the size of a set, $G = \{V, E\}$ denotes the GF graph, e_{ij} denotes the citation influence from paper v_i to v_j. The definition ensures that the source paper node v_i belongs only to the source topic T_k, not the destination topic T_l, consistent with the design of the GeneticScroll visualization.

In the chord diagram design, each arc in the circumference represents a topic, and each chord between two arcs represents the bidirectional flow between topics. Each arc is color-coded according to ColorBrewer's qualitative scales [11], and the chord applies a transition color from the source topic to the destination topic. The thickness of an arc visually encodes the total number of papers on that topic. The angle of each arc in the circular layout is proportional to the total efflux from the current topic to all other topics, as computed by Equation 1. The arc is further split into endpoints for all connecting chords to destination topics, with the circular angle of each endpoint proportional to the efflux from the current topic to each destination topic. The width of each chord smoothly transitions between the two endpoints. For example, the chord diagram in Figure 3(b) indicates that "classification" is the most prominent and influential topic. It is primarily influenced by, and exerts the greatest influence on, the topic of "DNN", with the bidirectional flow clearly highlighted.

Per-topic GF subgraph (Figure 3(c)) is visualized in the side view, using a hierarchical layout approach that maps the paper publication year to the vertical axis. This layout reveals the scientific research evolution of a scholar from a topic-based perspective. We employ the popular GraphViz library [20] with the Sugiyama-style hierarchical layout algorithm. To adapt this algorithm to our case, we bypass the first layer assignment step, as each paper node is already fixed to the layer corresponding to its publication year. The subsequent node reordering and coordinate computation steps are based on the original algorithm. After the layout, citation influence links among papers are drawn using third-order Bézier curves for aesthetic visualization, with additional dummy nodes used to ensure smoothness.

The GeneticPrism design is equipped with multimodal interactions. Initially, the GF graph data can be queried by multiple filters (Figure 1(a)) to control the number of nodes and edges, as well as the paper-topic affinity threshold. The 3D prism supports zoom&pan and can be rotated upon mouse drag to focus on topics more interesting to the current user. Another animation mode is also supported in that the prism will rotate around its center at a constant rate, presenting a dynamic overview for better analysis. Finally, double-clicking on a side view will drill down to the GeneticScroll visualization of the corresponding topic.

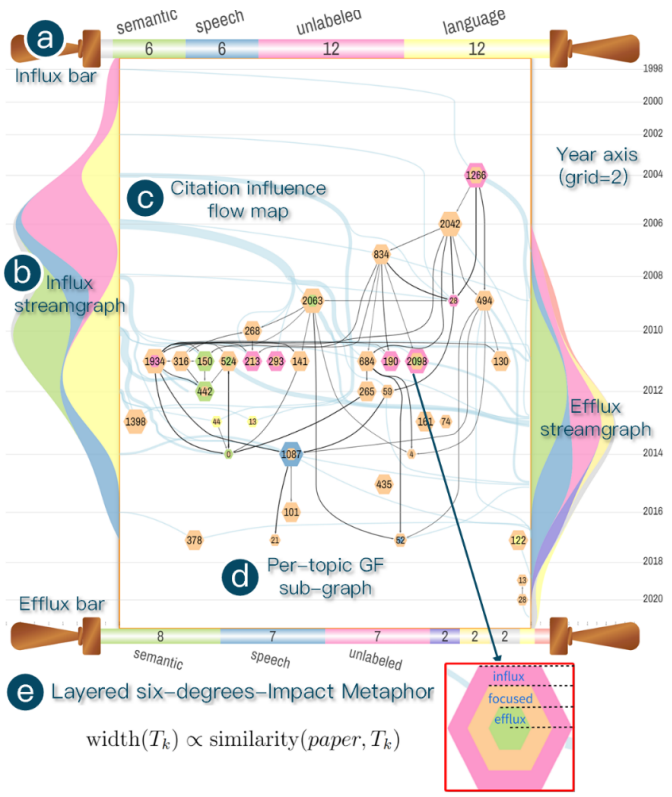


Fig. 4: GeneticScroll visualization for the topic "DNN" of scholar Andrew Y. Ng. It is composed of (d) a per-topic GF subgraph in the centre, (a) influx/efflux bars on the top and bottom sides, (b) influx/efflux streamgraphs on the left and right sides, and (c) citation influence flow maps in between, showing detailed inter-topic interactions. Each paper node is encoded with (e) a layered six-degrees-impact metaphor.

Alternative designs of GeneticPrism include several approaches that attempt to address the visualization tasks defined in Sec. III-D. Full GF graph (Figure 3(d)) represents the original GeneticFlow visualization, where all papers are displayed in a single graph. While this approach provides a broad overview, it fails to effectively address the complexity of the GF graph of highly productive scholars. As demonstrated in our User Evaluation, this design is prone to visual clutter in large graphs, making it difficult to interpret the research evolution on a single topic (T2) and inter-topic influence (T3). Force-directed layout (Figure 3(e)) is a non-hierarchical layout that ignores paper publication time, making it difficult to track the evolution of a scholar's research over time, which are in most cases hierarchically and temporally developed. Temporal-Thematic flow matrix (Figure 3(f)) uses time and topics as two axes, aggregating papers into matrix cells with the encoding of pie charts. The connections between pie charts represent the influence between topics. This approach attempts to visualize both the temporal evolution and topic interactions (T1 and T3), but it introduces overwhelming visual elements and edge crossings, making the overall structure difficult to interpret.

B. GeneticScroll

The main panel of our system will switch to the GeneticScroll view when the side view of GeneticPrism is focused and double-clicked. As shown in Figure 4, the GeneticScroll design consists of a hierarchical visualization of GF subgraph for the detailed analysis of a scholar's research evolution on a single topic (T2), with elaborate glyph designs on graph nodes to represent interdisciplinary papers (T4); two streamgraphs in the side view to display the topic dynamics influencing and influenced by the current topic; and the flow map in blue curved lines connecting between the streamgraphs and the GF subgraph to illustrate detailed topic interaction patterns surrounding the current topic (T3). The overall GeneticScroll design resembles a traditional scroll painting, highlighting the detailed and unfolding narrative of a scholar's research on a specific topic and its outreach.

The main component of GeneticScroll features the per-topic GF subgraph (Figure 4(d)) on the current topic, using the hierarchical layout with publication time as an axis. Since GeneticScroll is an expanded version of the side view, two design improvements are introduced to support an in-depth analysis of the impact and evolution of that topic. First, each paper node is now represented by a new glyph design of a layered six-degree-impact metaphor to highlight interdisciplinary papers. Second, the hierarchical node layout is optimized by a new algorithm to strike a balance between the layout athletics and the additional flow map visualization showing topic-level citation influences. The algorithm will be described in Sec. V.

Layered six-degree-impact metaphor for paper nodes, as enlarged in Figure 4(e), has a hexagon shape with the entire size visually encoding the class of high/medium/low citation number of the corresponding paper. The paper with more citations will be drawn in a larger hexagon. In its full version, the hexagon glyph has three layers, each representing a topic to which the current paper belongs. By this design, the paper with overlapping topics, i.e., the interdisciplinary work, can be visually highlighted (T4).

In detail, the middle layer of the glyph represents the focused topic of GeneticScroll, e.g., orange (topic DNN) in Figure 4(e). The outermost layer corresponds to another topic to which the current paper not only belongs but also receives an influx of citation influence from papers on that topic, e.g., pink (topic unlabeled). On the visualization, there should be at least one flow map link from the left-side streamgraph layer of that topic to the current paper node. In case there are multiple such topics, the topic with the highest similarity to the current paper is used. Similarly, the innermost glyph layer corresponds to the topic to which the current paper is sending efflux citation influence, e.g., green (topic semantic) in Figure 4(e). This layered design highlights the interdisciplinary nature of the paper, bridging multiple topics within the work by Andrew Y. Ng. As not all papers are necessarily interdisciplinary, the glyph degenerates into a one-layer or two-layer hexagon in most cases. If one of the layers of the glyph is empty, such as in the case of a two-layer hexagon, the design remains unambiguous because the middle layer is always present and has the same color corresponding to the current topic. The size of multiple layers on a single paper hexagon is determined in proportion to the paper's similarity to these topics, as calculated in Sec. III-C. Note that we do not display the external topic on a paper having no interaction with the

current topic/paper because this information will be visualized separately in the GeneticScroll view of that external topic.

Influx/efflux streamgraphs (Figure 4(b)) are presented on the left/right side of the GeneticScroll view, which displays the temporal dynamics of citation influence between other topics and the focused topic shown in GeneticScroll. For instance, the influx streamgraph on the left consists of multiple layers, each corresponding to a topic that contributes citation influence to the current topic. The height of each layer at a year indicates the total amount of influx citation influence from that topic to the focused topic in GeneticScroll in that year. For the "DNN" topic in Figure 4(b), the influx streamgraph highlights major influences from "unlabeled" between 2004–2008 and "language" between 2009-2014. Topic colors inherit those of GeneticPrism and GeneticScroll. In addition, on the top/bottom of GeneticScroll (Figure 4(a)), horizontal influx/efflux bars are designed to provide an overview of citation influence between external and focused topics. Each bar on the top/bottom indicates the influx/efflux of a topic, with the bar area proportional to the aggregated influx/efflux.

Citation influence flow map (Figure 4(c)) between influx/efflux streamgraphs and the GF subgraph visualization in the center reveals detailed topic-level interaction. The raw data are the citation influence links between the papers displayed in the current GF subgraph and the papers on the other topics. Visualizing these influence links simultaneously would result in significant visual clutter due to the GF subgraph already presented. We propose three approaches to resolve this issue. First, the raw citation influence links are aggregated into influx/efflux flows by the year granularity and connected to the edges of left/right-side streamgraphs. Flow thickness represents the strength of that flow. Second, the classical flow map layout [44] is adopted to arrange the flow visualization appropriately. We introduce a new algorithm in Sec. V to optimize the layout with the best effort and minimize edge crossings. Third, the influx/efflux flow map is rendered in light blue in the background, clearly distinguished from the GF subgraph links rendered in black in the foreground.

GeneticScroll visualization also supports several interactions to speed up the visual analysis. In addition to pan&zoom, clicking on nodes and links of the GF graph unfolds the details of selected papers and citation influence links in the information panel of Figure 1(e), including the paper title, abstract, venue, year, topics, and citation context, etc. Hovering an influx/efflux streamgraph layer highlights the part of the flow map representing the influx/efflux flows from/to the corresponding topic. Many other visualization parameters can also be visually configured, such as the year granularity for aggregating flows, the weight to layout GF graph vs. flow map, and the mode of edge bundling.

Alternative design of GeneticScroll (Figure 3(g)) features a GF subgraph with stacked bar charts. The per-topic GF subgraph is placed at the center, with stacked bar charts on the left and right representing citing and cited papers, respectively. Although this approach aims to visualize the influence and interaction among topics (T3), the layout treats the edges of internal citation and external influence separately, resulting in cluttered visualization, with severe edge crossings.

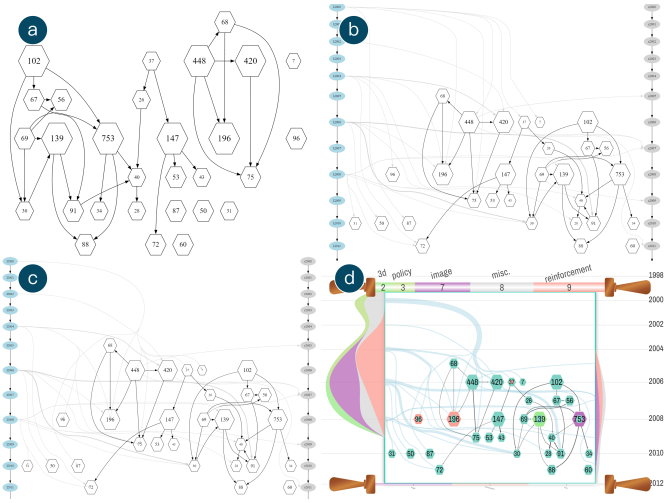


Fig. 5: Steps from the original GF sub-graph layout to the GeneticScroll view. (a) Original per-topic GF sub-graph; (b) GF sub-graph with influence edges; (c) GF sub-graph with influence edges using bundling; (d) GeneticScroll view: add flow map layout for influence edges and place them on the background layer, integrating with influx/efflux streamgraphs.

V. Integrated Flow Hierarchical Layout

We propose the Integrated Flow Hierarchical Layout (IFHL) algorithm for the GeneticScroll visualization, considering not only the layout of the central graph but also integrating its influence on other topics through flow map and streamgraph. In contrast to the designs in Figure 3(g), which directly draw citation influences over the per-topic GF sub-graph, our approach integrates the layout of the central graph and citation influences together. As shown in Figure 5, three steps are adopted to distinguish the central graph from the newly added influence flow map and effectively prevent visual clutter.

Objective of IFHL is to lay out the context edges integrated with the original hierarchical layout. The set of context edges includes all edges directly connected to topic T_k but not within T_k. The "in" edges (influx) connect to the upstream of the topic, while the "out" edges (efflux) connect to those downstream. We visualize these patterns by placing influx on the left and efflux on the right. This is achieved through influx nodes $v_l(y)$ and efflux nodes $v_r(y)$, where y is the year associated with the node. These nodes act as proxies for external endpoints: When a context edge connects from an external node to a topic-internal node, the source is replaced by the corresponding influx node; when connecting from an internal node to external, the target is replaced by the efflux node. This replacement yields influence edges, defined as:

$$\begin{aligned} E_I(T_k) = \big\{ e_{pj} \mid e_{ij} \in E,\ v_i \notin V(T_k) \wedge v_j \in V(T_k) \\ \rightarrow v_p = v_l(y(v_i)) \big\} \\ \cup \big\{ e_{ip} \mid e_{ij} \in E,\ v_i \in V(T_k) \wedge v_j \notin V(T_k) \\ \rightarrow v_p = v_r(y(v_j)) \big\} \end{aligned} \tag{2}$$

During GF graph layout, we adhere to the rules that influx nodes are always positioned on the left side and efflux nodes on the right side, arranged chronologically by year y. Influx nodes v_l are placed at the extreme left and efflux nodes v_r

at the extreme right of the layout. We enhanced GraphViz's Hierarchical Layout to process the whole graph with influence edges and proxy nodes.

Weighted edge crossing is adopted as the optimization objective rather than edge crossing. In the objective function, we use $\text{cost}(e)$ instead of the default "1" as the edge weight, defined as the edge's own width if it belongs to $E_I(T_k)$, and α otherwise. The weighted crossing between e_1 and e_2 is defined as $\text{cost}(e_1) \cdot \text{cost}(e_2)$. We use total weighted crossing as the optimization objective. To evaluate the optimal weight α for the best layout, we use crossing counts as indicators, including internal crossing (between central edges) and external crossing (between influence edges). We aim to find an optimal objective where the number of internal crossings is low (primary goal) while maintaining low external crossings. Finally, the elbow method is introduced to determine the optimal α. Our algorithm calculates an adaptive α, resulting in an enhanced topic GF graph with detailed influence information and minimal deviation from the original layout.

Citation influence flow map in GeneticScroll employs a two-step process to achieve progressively widening edges at intersections (mimicking Sankey diagrams), minimizing clutter and crossings (Figure 5(d)). First, edge bundling merges adjacent dummy nodes specifically for influence edges, reducing crossings and compacting the layout while preserving the central graph structure (Figure 5(c)). Then, flow map adjustment models these bundled intersections, refining edge positions and progressively increasing their thickness to achieve the final Sankey-style flow. This method results in a clear visualization integrating the central graph and influence flows.

Based on both theoretical complexity and empirical benchmarks, our proposed IFHL algorithm exhibits quadratic runtime scaling ($O(|V|^2)$), as evidenced by the log-log scalability analysis yielding a slope of 2.16 in Appendix 2.4. Although the worst-case theoretical complexity ranges from $O(|V||E|)$ to $O(|V|^3)$, our graph partitioning strategy ensures that the majority of domain-specific subgraphs (at the 95th percentile) contain fewer than 135 nodes. As a result, the algorithm achieves execution times under 178 ms per subgraph, thereby enabling real-time interaction in standard web browsers via WebAssembly. Further details regarding the algorithm and complexity analysis are provided in Appendix 2.

VI. Case Studies and User Evaluation

GeneticPrism is publicly available for exploration and use. The web application (https://genetic-flow.com) provides immediate, interactive access to the Research Evolution Graphs for all scholars analyzed in the following case studies. All source code, full datasets, deployment instructions, a user manual, and a demo video are hosted in the open-source repository (https://github.com/visdata/GeneticPrism); user feedback is welcome via its issue tracker.

A. Prof. Hanrahan's Interdisciplinary Impact

Prof. Pat Hanrahan is an ACM Turing award laureate (2019) known for fundamental contributions to 3D computer graphics and computer-generated imagery [5]. The first case study leverages GeneticPrism to visually analyze Hanrahan's research evolution, focusing on his contributions to interdisciplinary research topics.

Initially, using the original GF graph visualization (Figure 6(a)), we identify three primary research topics associated with Hanrahan's work: 3D graphics (orange), parallel hardware (green), and visualization (pink). However, the three topics somehow intertwine over time, and it is challenging to see the evolution structure of each topic clearly. Though the topic interaction can be observed through links between differently colored nodes, since each paper is only encoded as one topic, it is also impossible to identify key interdisciplinary works and their impact enclosure.

By the new GeneticScroll view, Prof. Hanrahan's research on the 3D graphics topic over 30 years can be summarized as a single picture of Figure 6(b). Notably, the two pioneering papers enclosed within the red box are easily identifiable, as they occupy the top positions in the graph and serve as the source of many subsequent nodes. These papers are the RenderMan shading language published in 1990 (193 citations) and the hierarchical radiosity algorithm from 1991 (506 citations). Hanrahan's later works in 3D graphics primarily branch out from these papers, forming two main research threads: one on shading languages and the other on textures and radiosity.

From the early 1990s, as his Turing award statement also mentions, the 3D graphics topic gradually incubated another topic alongside, as evidenced by the green layer in the right-side streamgraph (Figure 6(b)). This is the parallel hardware topic, which mainly refers to the use of GPUs to accelerate graphics processing. The total efflux from 3D graphics to parallel hardware amounts to 12 extensions. The topic interactions are featured by two influential, interdisciplinary papers enclosed within the black box, all having colored layers on both topics. The first SIGGRAPH'01 paper (194 citations) describes a shading system on new graphics hardware, and the second SIGGRAPH'02 paper (577 citations) deploys ray tracing algorithms to the GPU hardware.

Figure 6(c) gives an overview of Hanrahan's work on the parallel hardware topic. His contributions there can be roughly clustered into two sub-topics: GPU-based high-performance computation for graphics and advanced graphic programming models and compilers. Interestingly, the topic again triggers a new topic "visualization", as shown by the pink-layer streamgraph in Figure 6(c), with a topic efflux of 15 extensions. Three papers close to the pink layer contribute the most to this transition of research. Two of these papers are interdisciplinary between parallel hardware and visualization, as shown by the multiple color layers. After mouse hovers to access their titles, these researches are found to be the visualization of parallel hardware's performance and application behavior. We further drill down to the citation influence links from the hardware research to visualization. The citation contexts in Figure 6(d) reveal a key clue on how they developed the famous Polaris visualization system: researchers first invented Argus [32] (a parallel graphics library, one of the three papers, 73 citations), but found scalability issues on its performance, so they decided to apply visualization tools to analyze the performance [8] (another of the three papers, 36 citations).

In Figure 6(e), the evolution of Hanrahan's visualization

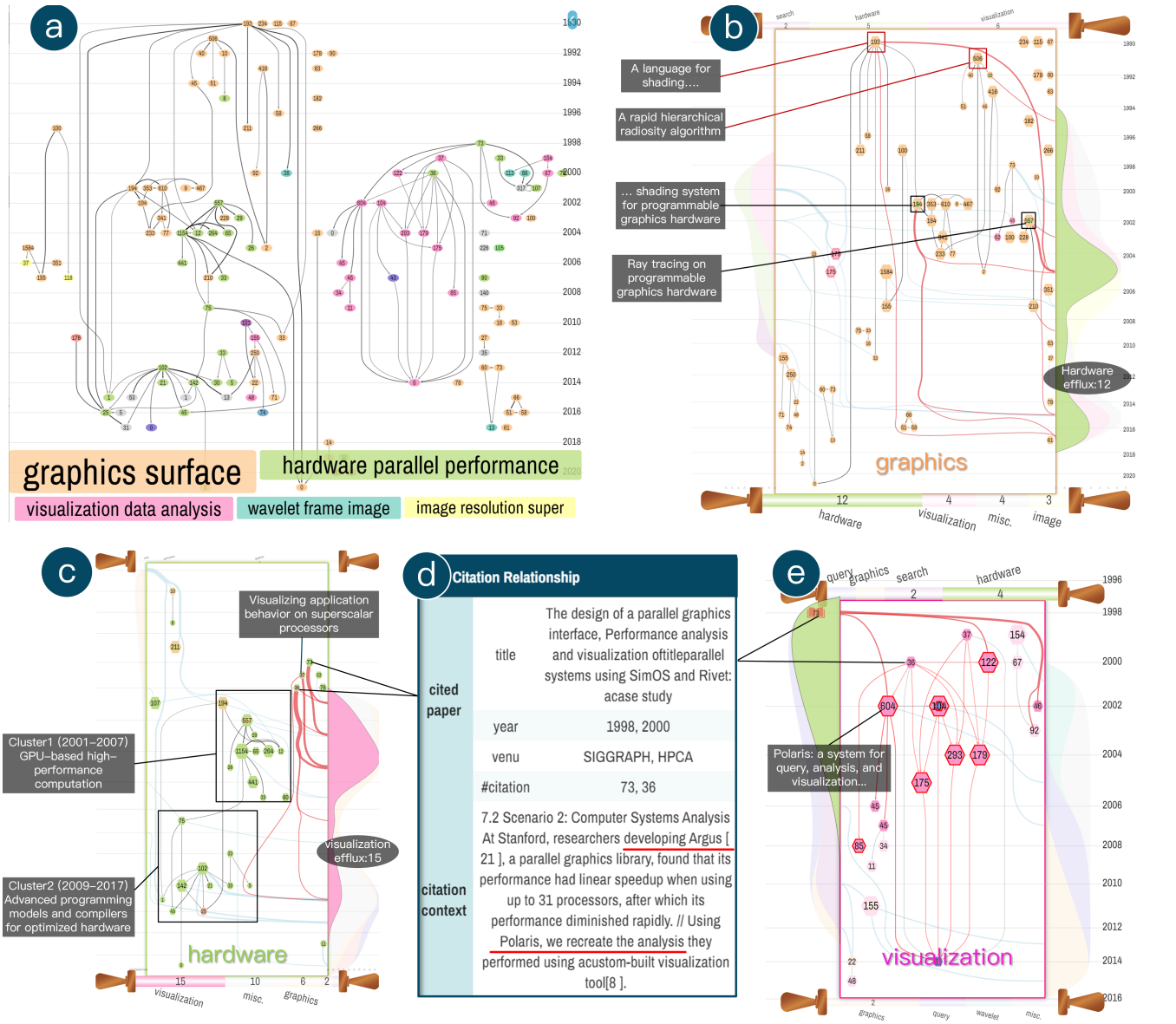


Fig. 6: Case study on Pat Hanrahan's interdisciplinary impact: (a) full GF graph; (b) GeneticScroll view on the topic "3D graphics"; (c) GeneticScroll view of topic "parallel hardware"; (d) detailed citation information of paper "Polaris"; (e) GeneticScroll view on the topic "visualization".

work is illustrated. By examining all paper titles, it can be found that the primary component is seven publications focusing on the Polaris visualization system, with three papers, three patents, and one case study about the system, as indicated by the nodes in the red outline. Notably, one KDD'02 paper [51] (104 citations) within them is shown to be interdisciplinary with database research. A further drill-down into the details explains this finding as the application of Polaris to the analysis of multidimensional databases. It is indeed well-known that Polaris/Tableau's initial customers are from the database community.

In a few visual analysis walks with GeneticPrism, we unveil the three major contribution areas of Prof. Hanrahan, as well as his proven stories on topic evolution and transition. The case strongly demonstrates the effectiveness of our visualization technique in illustrating and explaining both intra-topic and inter-topic research evolution on individual scholars.

B. Prof. Stonebraker's Lifelong Database Research

In the second case study, we apply GeneticPrism to understand the research evolution of Prof. Michael Stonebraker, another renowned ACM Turing award laureate (2014). Prof. Stonebraker has significantly influenced the field of databases, and his pioneering works have laid the foundation for many modern database technologies.

Figure 1 provides a topic overview of Prof. Stonebraker's research by the chord diagram in GeneticPrism. We observe two primary research topics: database (orange) and storage (green), along with several minor research topics such as visualization (pink) and cloud. The database topic dominates his research with 107 core papers out of all 174 papers (61.5%). The GeneticPrism view (Figure 1(d)) putting his two major topics side by side echoes the dominance of the database topic over the entire period and exposes the richness and connectedness of his research on the topic. In comparison,

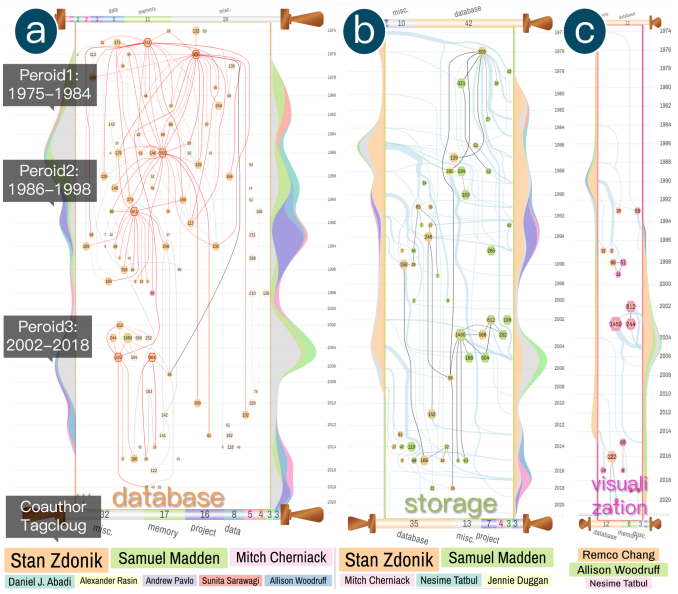


Fig. 7: Case study on Michael Stonebraker's academic career: (a)(b)(c) GeneticScroll visualization of "database", "storage", and "visualization" topics, respectively.

the storage topic is featured only in a few research threads with tens of publications.

Drilling down to the GeneticScroll panel for details on the database topic, Figure 7(a) sketches a global view showing densely interrelated research. The hierarchical node-link graph drawing attached to the time axis identifies three major periods of Prof. Stonebraker's database research, each with a few landmark papers influencing tens of follow-up papers to form a thread. In the first period (1975-1984), two papers with 411 and 805 citations (in red outline) introduce the INGRES system. In the second period (1986-1998), the two papers with 665 and 401 citations developed the Postgres system. The third period (2002-2018) features two papers with 325 and 964 citations, both studying the latest commercial Database Management System (DBMS). These discoveries correspond precisely to the three contributions in Prof. Stonebraker's Turing award statement [4]: INGRES, Postgres, and Entrepreneurship on DBMS. The influences of these six key papers (highlighted in red) almost cover the entire landscape of his database research.

We further expand the GeneticScroll views on the "storage" and "visualization" topics (Figure 7(b) and Figure 7(c)). Unlike the database view, these two topics share a key characteristic: their graphs are characterized by widely distributed orange nodes, representing interdisciplinary research with the "database" topic, rather than solely by works from the current topic. This pattern is consistent across other minor topics, suggesting that most of Prof. Stonebraker's works in these areas focus on applying database technology or providing tool support (e.g., for visualization). From the coauthor tag clouds at the bottom of Figure 7, we observe that Prof. Stonebraker's collaborators for "database" and "storage" topics are highly consistent, with the top three coauthors being identical. In contrast, the "visualization" topic features a different set of collaborators, indicating that "database" and "storage" are

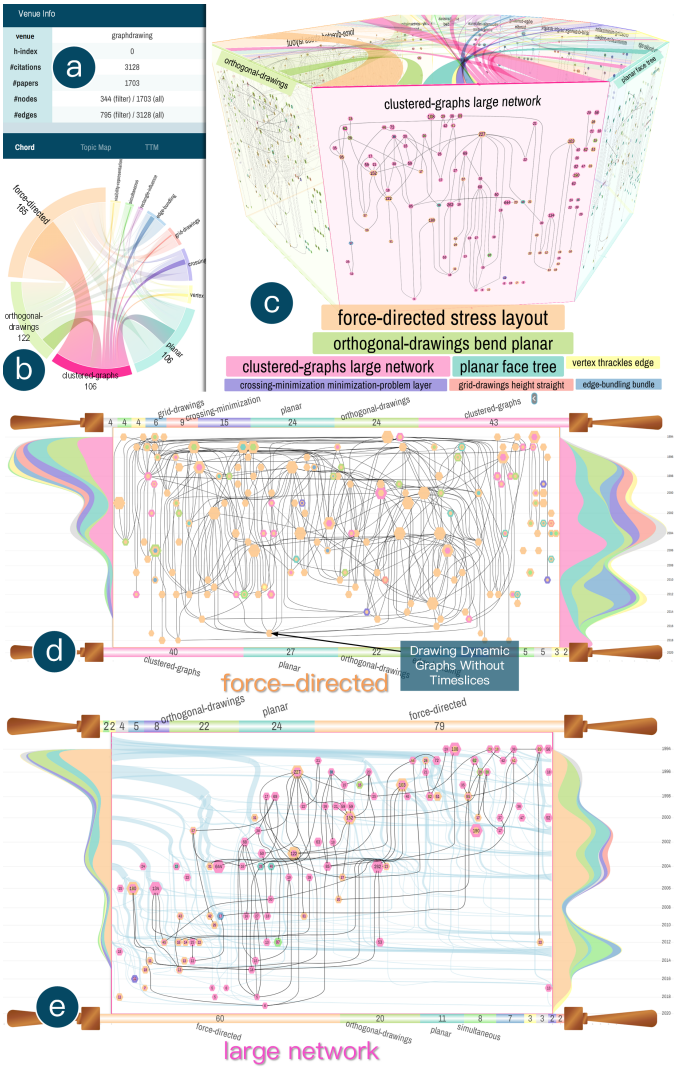


Fig. 8: Case study on the GD symposium: (a) scholar demographics; (b) chord diagram; (c) GeneticPrism visualization; (d)(e) GeneticScroll visualization of "force-directed", "clustered-graphs" topics, respectively.

closely related topics, while "visualization" is less connected. These GeneticScroll visualizations clearly outline Prof. Stonebraker's lifelong dedication to database research.

C. The Backbone and Outreach of Graph Drawing Symposium

Beyond the above cases on prestigious scholars, the GeneticPrism technique can also be applied to understand the evolution of academic events, e.g., International Symposium on Graph Drawing and Network Visualization (GD) [3]. We extract all the GD papers from MAG (1994∼2021), up to 1703 papers. To accommodate this large data in GeneticPrism, only high-impact papers are presented, filtering by their number of citations: $\geq$ 5 for papers newer than 2017, $\geq$ 10 for papers newer than 2012, and $\geq$ 15 for other papers. The final GD dataset has 344 papers (nodes) and 795 citations (edges).

As illustrated in Figure 8(b), these high-impact GD papers form 11 topics, with a single backbone topic of "force-directed" having 165 papers, covering half of all GD papers. Three other topics displayed in Figure 8(c) are the second

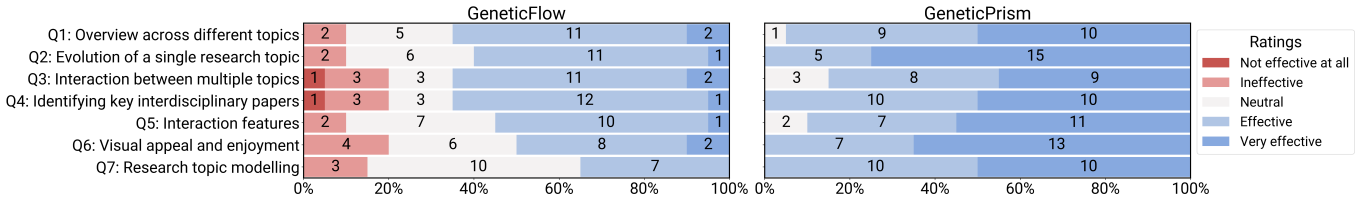


Fig. 9: Comparison of user ratings for GeneticFlow and GeneticPrism across seven evaluation criteria (Q1-Q7). The bar lengths reflect the distribution of responses, with darker shades representing higher perceived effectiveness.

largest (also indicated by the arc thickness): "orthogonal-drawings" published before 2014 (with over 90% papers), "planar drawing and trees", and the "clustered-graph and large network visualization" topic studied in the whole time of GD.

Drilling down to the force-directed topic with GeneticScroll reveals its backbone nature for GD. In the densely connected graph of Figure 8(d), more than half of all nodes are encoded with multiple color layers, representing interdisciplinary research applying force-directed methods to various graph drawing problems. Meanwhile, it is observed that many of the highest-impact papers drawn in the largest hexagons are single-colored. This suggests that the inherent advancement of force-directed methods still serves as a major scientific evolution in the GD symposium. The GeneticScroll view of the large network visualization topic with lasting impact presents a major outreach of GD (Figure 8(e)). The topic was initially influenced by force-directed methods. After that, new requirements in visualizing large graphs in return drove the development of force-directed drawing, as shown by the streamgraph in the right part of Figure 8(e). In the recent years of 2010∼2016, it even triggered a new research topic on "simultaneous embedding" (in green color), which lays out multiple graphs simultaneously to solve the challenge of multi-level or multi-component drawing in visualizing large graphs.

D. User Evaluation

We evaluated GeneticPrism through an in-lab user study with 20 participants from various backgrounds and data analysis expertise. Specifically, we focused on research tasks defined in Sec. III-D. To quantitatively evaluate GeneticPrism's effectiveness, we designed a questionnaire to measure participants' ratings on system effectiveness and usability. The full version of the questionnaire is included in Appendix 3, while the short version is located in Figure 9. We recruited 20 participants (16 males, and 4 females) through social media and word-of-mouth. The participants came from diverse academic fields, including databases, machine learning, visual analytics, industrial engineering, computational sociology, library science, and HCI. For comparison, we used GeneticFlow [60], a previous system designed to visualize a scholar's research through self-citation graphs using a time-dependent, hierarchical representation method, as the baseline.

Figure 6 compares the distribution of user ratings for the baseline and GeneticPrism based on our self-developed questionnaire. GeneticPrism consistently achieved higher median ratings across task performance (Q1-Q4), interaction effectiveness (Q5), user engagement (Q6), and topic modeling effectiveness (Q7), with 95.7% of responses rated above neutral. On a 5-point Likert scale (1: not effective at all, 5: very effective), participants found that in comparison to GeneticFlow, using GeneticPrism made it easy to understand the evolution of a single research topic, (Med = 5.0, M = 4.5, SD = 0.44), interact with the diagram (Med = 5.0, M = 4.45, SD = 0.69), and considered it visually appealing and enjoyable (Med = 4.5, M = 4.65, SD = 0.49). As the sample size is relatively small, we performed the Wilcoxon signed-rank test [56], which showed significant differences ($p < 0.001$) for all questions.

In terms of understanding a scholar's scientific evolution (Q1), 10 participants noted that GeneticPrism's multidimensional representation effectively separated research topics across distinct facets, making thematic evolution easier to observe. P3 mentioned, "The 3D prism clearly separates different interests, helping evaluate how the author's thought evolves, while scroll presents topic elements without distractions from unrelated nodes." For interaction between topics (Q3), 9 participants praised the combined views of GeneticScroll's timeline and the chord diagram's global overview for conveying topic relationships. P5 noted, "The chord diagram provides a quick overview of citation patterns, while GeneticScroll shows detailed flow over time." In identifying interdisciplinary works (Q4), 10 participants emphasized GeneticPrism's ability to encode cross-disciplinary influences through color-coded nodes and layers. P3 stated, "The hexagonal node layers intuitively display cross-disciplinary attributes of key papers." In contrast, 7 participants found GeneticFlow's single-color nodes less effective for this purpose. In terms of visual design and enjoyment (Q6), 13 participants praised GeneticPrism's layered, multidimensional design and vibrant color schemes. P4 remarked, "The 3D prism metaphor feels like a monument that visually represents the scholar's research legacy." While P14 mentioned, "Without clear labels, the 3D view can feel overwhelming." For multi-topic modeling (Q7), all participants agreed that GeneticPrism's approach captured interdisciplinary studies more effectively by assigning multiple topics to a paper. P6 commented, "Many studies overlap across fields, and the multi-topic approach captures this complexity better."

In summary, our user evaluation with 20 participants evaluated GeneticPrism on 7 questions, show significant improvements in task performance, interaction, topic modeling, and user engagement, with 95.7% of responses rated above neutral. In open-ended questions, despite the learning curve for the 3D prism design, participants praised the 3D prism for effectively separating research topics, the scroll for analyzing inter-topic relationships and citation patterns, the hexagonal node design for encoding cross-disciplinary influences, and the multi-topic modeling for capturing research complexity.

VII. Discussion

Through the case study, we demonstrate that GeneticPrism visualization can effectively delineate the major research threads of top scholars and academic venues and help analyze both intra-topic and inter-topic impact evolutions. Compared with previous visualization methods, the advantage lies in the strategy of slicing one's citation influence graph into different topic facets. This not only reduces the visual complexity in each single view but also allows us to visually analyze the impact evolution of each topic independently.

Meanwhile, a potential limitation of GeneticPrism is its customized design for the self-citation influence graph of individual scholars. This characterizes the current study as application-oriented. However, efforts to extract a generic methodology from this approach reveal that GeneticPrism can be adapted to visualize general multivariate networks, where a primary node attribute serves as the topic attribute. The underlying assumption is that this node attribute is nominal or categorical; however, with appropriate preprocessing, numerical or ordinal attributes can also be accommodated. Consequently, the GeneticPrism approach is effective for multivariate networks in which nodes sharing the same attribute value are positioned more closely than those with differing values. Such conditions are commonly met in practice—in extreme cases, graph clustering can offer cluster-ID as the topic attribute.

Another potential future work would be extending GeneticPrism to support the analysis of multiple scholars in the same group or community. The single hierarchy design from GeneticPrism to GeneticScroll can be upgraded to having multiple hierarchies, with the scholar facets at the top and more topic facets at the bottom. To reduce the visual complexity in the GF graph of each scholar, network abstraction/aggregation techniques should be studied carefully. Since the number of topics involving multiple scholars will increase a lot, hierarchical topic modeling can also be introduced to create multiple levels in the topic facet visualization of GeneticPrism.

VIII. Conclusion

This work studies an essential problem of visualizing the scientific research evolution of individual scholars from multiple topic facets. Building on the previous work of GeneticFlow, we propose an end-to-end technical framework that includes academic data curation and pre-processing, citation influence graph analysis and topic modeling, and the GeneticPrism visualization designs and interactions, which are our main contributions. A new 3D prism-shaped visual metaphor is introduced for research evolution overview, while a GeneticScroll design is proposed to orchestrate both intra-topic temporal/hierarchical research evolution and inter-topic citation interactions, with an elaborate layout algorithm to minimize edge crossing and visual clutter. The visualization techniques are demonstrated to be effective through three case studies in delineating the intriguing academic lives of Turing award laureates and the evolution of the GD symposium. A focused user evaluation further validates the system, showing significant improvements in task performance, interaction, topic modeling, and user engagement, with 95.7% of responses rated above neutral.

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